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EXAMINER	
MALKOWSKI, KENNETH J	

ART UNIT	PAPER NUMBER
3667	

NOTIFICATION DATE	DELIVERY MODE
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Please find below and/or attached an Office communication concerning this application or proceeding.

The time period for reply, if any, is set in the attached communication.

Notice of the Office communication was sent electronically on above-indicated "Notification Date" to the following e-mail address(es):

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Office Action Summary	Application No. 17/691,475	Applicant(s) Ascherl et al.	
	Examiner KENNETH J MALKOWSKI	Art Unit 3667	AIA (FITF) Status Yes

-- The MAILING DATE of this communication appears on the cover sheet with the correspondence address --

Period for Reply

A SHORTENED STATUTORY PERIOD FOR REPLY IS SET TO EXPIRE 3 MONTHS FROM THE MAILING DATE OF THIS COMMUNICATION.

- Extensions of time may be available under the provisions of 37 CFR 1.136(a). In no event, however, may a reply be timely filed after SIX (6) MONTHS from the mailing date of this communication.
- If NO period for reply is specified above, the maximum statutory period will apply and will expire SIX (6) MONTHS from the mailing date of this communication.
- Failure to reply within the set or extended period for reply will, by statute, cause the application to become ABANDONED (35 U.S.C. § 133). Any reply received by the Office later than three months after the mailing date of this communication, even if timely filed, may reduce any earned patent term adjustment. See 37 CFR 1.704(b).

Status

1)☑ Responsive to communication(s) filed on 11/20/24.

☐ A declaration(s)/affidavit(s) under **37 CFR 1.130(b)** was/were filed on _____.

2a)☑ This action is **FINAL.** 2b) ☐ This action is non-final.

3)☐ An election was made by the applicant in response to a restriction requirement set forth during the interview on _____; the restriction requirement and election have been incorporated into this action.

4)☐ Since this application is in condition for allowance except for formal matters, prosecution as to the merits is closed in accordance with the practice under *Ex parte Quayle*, 1935 C.D. 11, 453 O.G. 213.

Disposition of Claims*

5) ☑ Claim(s) 1-4,6-17 and 20 is/are pending in the application.

5a) Of the above claim(s) _____ is/are withdrawn from consideration.

6) ☐ Claim(s) _____ is/are allowed.

7) ☑ Claim(s) 1-4,6-17 and 20 is/are rejected.

8) ☐ Claim(s) _____ is/are objected to.

9) ☐ Claim(s) _____ are subject to restriction and/or election requirement

* If any claims have been determined allowable, you may be eligible to benefit from the **Patent Prosecution Highway** program at a participating intellectual property office for the corresponding application. For more information, please see http://www.uspto.gov/patents/init_events/pph/index.jsp or send an inquiry to **PPHfeedback@uspto.gov.**

Application Papers

10)☐ The specification is objected to by the Examiner.

11)☐ The drawing(s) filed on _____ is/are: a)☐ accepted or b)☐ objected to by the Examiner.

Applicant may not request that any objection to the drawing(s) be held in abeyance. See 37 CFR 1.85(a).

Replacement drawing sheet(s) including the correction is required if the drawing(s) is objected to. See 37 CFR 1.121(d).

Priority under 35 U.S.C. § 119

12)☐ Acknowledgment is made of a claim for foreign priority under 35 U.S.C. § 119(a)-(d) or (f).

Certified copies:

a)☐ All b)☐ Some** c)☐ None of the:

1.☐ Certified copies of the priority documents have been received.

2.☐ Certified copies of the priority documents have been received in Application No. _____.

3.☐ Copies of the certified copies of the priority documents have been received in this National Stage application from the International Bureau (PCT Rule 17.2(a)).

** See the attached detailed Office action for a list of the certified copies not received.

Attachment(s)

1) ☐ Notice of References Cited (PTO-892)

2) ☐ Information Disclosure Statement(s) (PTO/SB/08a and/or PTO/SB/08b) Paper No(s)/Mail Date _____.

3) ☐ Interview Summary (PTO-413) Paper No(s)/Mail Date _____.

4) ☐ Other: _____.

DETAILED ACTION

Notice of Pre-AIA or AIA Status

The present application, filed on or after March 16, 2013, is being examined under the first inventor to file provisions of the AIA.

Response to Amendment

The amendment dated 11/20/24 has been accepted and entered. Accordingly, claims 10 and 15 are amended and claims 5 and 18-19 are canceled. Accordingly, claims 1-4, 6-17 and 20 are examined herein.

Claim Objections

Claim objection to claim 10 has been withdrawn as a result of the amendment.

Response to Arguments

With respect to the rejection of claim 1, Applicant asserts the combined teachings of Ogura, Gol, Foster, Ikenori and Fuji fail to teach, disclose or suggest "wherein the processor is configured to selectively cause the autonomy approval function to display a depiction of the autonomous work machine with a visual indication that the autonomous work machine is initiating motion" (Amend. 2).

Applicant reasons this is because Fuji does not provide a teaching or suggestion of "a depiction of the autonomous work machine" (Amend. 2). However, claim 1 is a rejection in view of Ogura, Gol, Foster, Ikenori and Fuji.

For example, Ogura discloses an autonomy approval function depicting an autonomous work machine (¶¶ 84-85 "display device 113 . . . work state display area 113C . . . travel route R, an actual position, worked position and the like can be displayed"; 113C depicting icons of an autonomous work machine actual position, i.e., in FIG. 4-7, 9, etc.; 90 the remote control area 113A and the work state display area 113C may be displayed so that the autonomously moving

work vehicle 1 can be operated while the work position is checked."). In addition, Gol (2, FIG. 6 and corresponding description; 12, FIG. 3-5 and corresponding description), Foster (68-72, FIG. 3 and corresponding description) also disclose a depiction of an autonomous work machine

Fuji is primarily cited to modify the disclosures of the combined teachings Ogura, Gol, Foster, and Ikenori such that the displayed autonomous work machine is further provided with a visual indication the autonomous work machine is initiating an autonomous motion. For example, element 51 in FIG. 5-7 is a countdown that is a visual indication the autonomous work machine is initiating an autonomous motion. Element 251 in FIG. 14-16 is also a visual indication the autonomous work machine is initiating an autonomous motion. These disclosures render the disputed language obvious in view of the combined teachings of Ogura, Gol, Foster, Ikenori and Fuji.

Furthermore, the arrow 355 shown in FIG. 18 with display 314a/314b is a display of a depiction of the autonomous vehicle indicating when the AV will be initiating a motion as indicated by the position of the start of arrow 355 and as also indicated by countdown display 351. Arrow 355 is also shown in FIG. 8 (D) with vehicle A and its corresponding representation indicated by the arrow emerging from vehicle A (¶¶ 112-116). Accordingly, Fuji alone additionally teaches "display a depiction of the autonomous work machine with a visual indication that the autonomous work machine is initiating motion". Accordingly, Applicants' arguments are unpersuasive.

Claim Rejections - 35 USC § 103

In the event the determination of the status of the application as subject to AIA 35 U.S.C. 102 and 103 (or as subject to pre-AIA 35 U.S.C. 102 and 103) is incorrect, any correction of the statutory basis (i.e., changing from AIA to pre-AIA) for the rejection will not be considered a new

ground of rejection if the prior art relied upon, and the rationale supporting the rejection, would be the same under either status.

The following is a quotation of 35 U.S.C. 103 which forms the basis for all obviousness rejections set forth in this Office action:

> A patent for a claimed invention may not be obtained, notwithstanding that the claimed invention is not identically disclosed as set forth in section 102, if the differences between the claimed invention and the prior art are such that the claimed invention as a whole would have been obvious before the effective filing date of the claimed invention to a person having ordinary skill in the art to which the claimed invention pertains. Patentability shall not be negated by the manner in which the invention was made.

Claims 1-3, 10, 13-14 are rejected under 35 U.S.C. 103 as being unpatentable over US 20170248946 to Ogura et al. (Ogura) in view of US 20220161783 to Golgiri et al. (Gol) and further in view of US 20170357400 to Foster et al. (Foster) and further in view of US 20210362718 to Ikenori (Ikenori) and further in view of US 20180297611 to Fujisawa et al. (Fuji)

With respect to claim 1, Ogura discloses a mobile device comprising:

a processor;

memory, coupled to the processor, the memory containing instructions that when executed cause the processor to provide an autonomy management application, wherein the autonomy management application includes an autonomy approval function;

(¶ 31 remote control apparatus 112; 51 remote control apparatus 112 . . . can be taken out from the auxiliary moving work vehicle 100 and operated while being carried . . . tablet type personal computer; 52 remote control apparatus 112 and the autonomously moving work vehicle 1 can be communicated with each other . . . display device 113 which is a touch panel-type operation screen which can be operated by touching the screen is provided; 55 operator can operate the remote control apparatus 112 while supervising the autonomously moving work vehicle 1; 56-57 remote control area 113A is a screen in which operation members (buttons) for travel, steering and operation of the work machine of the autonomously moving work vehicle 1

are arranged . . . automatic travel start button 203, an automatic travel stop button 204, a sudden stop button 205; ¶¶ 64-67)

the autonomy approval function providing a user interface configured to display an autonomy job[1] to a user, the display including a map of a selected field, the user interface being configured to receive user acceptance of the autonomy job;

(64 automatic travel start button 203 . . . button for starting work while traveling automatically on the travel route R set previously . . . shifted to automatic travel mode . . . automatic travel control is performed by the control device 30)

(i.e., map in work state display area 113c shown in various figures, i.e., 4 and corresponding descriptions; ¶ 53 the set travel route R is set previously in the field H as shown in Fig. 3)

(mobile device 112, display 113 shown in FIG. 2, 4-12 and corresponding description; ¶¶ 57 -59 remote control area 113A, screen in which operation members (buttons) for travel, steering, an operation of the work machine of the autonomously moving work vehicle are arranged with various examples; 60-67; 71-74);

the autonomy approval function including a begin mission user interface element configured to receive user input and wherein the processor is configured to selectively cause the autonomy approval function to initiate motion based on the user input to the begin mission user interface element

(i.e., ¶¶ 56-57 remote control area 113A is a screen in which operation members (buttons) for travel, steering and operation of the work machine of the autonomously moving work vehicle 1 are arranged . . . automatic travel start button 203, an automatic travel stop button 204, a sudden stop button 205; ¶¶ 60-67; ¶¶ 57 -59 remote control area 113A, screen in

[1] No limiting definition or examples are provided for “autonomy job” or an “aspect” thereof.

which operation members (buttons) for travel, steering, an operation of the work machine of the autonomously moving work vehicle are arranged with various examples)

However, Ogura fails to explicitly disclose the autonomy approval function being configured to instruct the user to move within a pre-determined distance from of an autonomous work machine that will execute the autonomy job, and provide a machine-verified indication of user proximity within the pre-determined distance such that the begin mission user interface is provided based on proximity of the user within the predetermined distance, i.e., an ability to select initiation of an autonomous motion can begin once the user is a predetermined distance from the work machine.

GoI is from the same field of endeavor, as GoI also discloses a remote device (i.e., 14, FIG. 1) wherein the remote device, like Ogura can be a tablet with a display (¶ 37 "interface on a remote device . . . smart phone . . . tablet") which allows a user to control motions of a vehicle in an unmanned manner (¶ 38 control the motive functions of the vehicle . . . without a driver occupying the vehicle) wherein GoI further discloses an autonomy approval function being configured to instruct the user to move within a pre-determined distance from of an autonomous work machine that will execute an autonomy job, and provide a machine-verified indication of user proximity within the pre-determined distance such that an automated job can only be performed after verification that a user has moved within the pre-determined distance of the autonomous work machine to initiate motion.

(¶¶ 37 system autonomously parks the vehicle 12 . . . user U may activate the remote parking system 10 using a button on a remote device 14 . . . smart phone . . . tablet, etc.; 41-42 remote device 14, the wireless communication circuit 24 facilitates communication between the user U and the system 10 (e.g., activating the remote parking system, etc.) . . . detect the location of the user U based on the range detection sensors . . . detect a location of the remote device 14 to position the user U . . . within a predetermined range of the vehicle 12 . . . as a

condition to activating the remote parking routine; 48 require user is in viewing zone to initiate autonomous operation of the vehicle)

(¶ 37 interactive routine . . . instruct . . . a position of the user U within a viewing zone . . . guidance routine to position a user . . . in the viewing zone; 40 including references incorporated by reference; 49 user instructed to relocate within a predetermined distance, i.e., within viewing zone; 52-53, FIG. 4-6 including instructions on remote device; FIG. 8 142 "outputting instructions coaching the user to enter a viewing zone").

Accordingly, it would have been obvious to one of ordinary skill in the art at the time of effective filing date to implement the teachings of GoI into the system of Ogura such that a begin mission user interface element (i.e., Ogura start 203, FIG. 6, 203A, FIG. 9) is displayable after verifying a user has moved withing a predetermined distance of an autonomous vehicle, as taught by GoI, since GoI discloses user proximity to the vehicle is a precondition to a user being able to approve initiation of motion in order to improve the ability of a user to monitor autonomy functions and in order to improve safety by ensuring the user does not interfere with the movement of the autonomous vehicle while also allowing the user to monitor operations of the AV, i.e., ¶ 1 "assisting a user-pedestrian monitoring the operation of a vehicle"; 37 the system still may benefit from pedestrian-user observation and monitoring . . . optimized viewing).

In addition, Ogura in view of GoI fail to explicitly disclose the display map includes displaying *pathing information* relative to the selected field, although Ogura at least suggests this feature (i.e., map in work state display area 113c shown in various figures, i.e., 4 and corresponding descriptions; ¶ 53 the set travel route R is set previously in the field H as shown in Fig. 3).

However, displaying pathing information along with map information was well known in the art at the time of effective filing. For example, Foster, from the same field of endeavor, discloses a user interface display (FIG. 3 and corresponding description) with a mapped grid of a field (right side, FIG. 3 including 74 (¶ 29 "agricultural field 74)) that includes pathing

information (74, 76, FIG. 3 and corresponding description, ¶28 "controller may plan a path 76 for the agricultural vehicle . . . to traverse"). In addition, Foster discloses includes an autonomy approval function on the user interface to receive user acceptance of the job including a begin mission user interface element receiving user input to selectively cause initiation of motion (¶29 "controller may then receive a third input from the operator indicative of instructions to commence an autonomous agricultural operation on the agricultural field . . . touchscreen display 34 indicative of instructions to display a start button 78 while the virtual agricultural vehicle 70 and/or implement 72 and the virtual agricultural field 74 are selected. The controller may receive signal(s) from the display indicative of actuation of the start button 78 and may commence the autonomous agricultural operation accordingly (e.g., by sending signal(s) to the steering control system and/or the speed control system indicative of instructions to direct the agricultural system along the path; 37; 39; claim 13).

Accordingly, it would have been obvious to one of ordinary skill in the art at the time of effective filing date to include pathing information in the user display as disclosed by Foster above, in the system of Ogura in view of GoI in order to improve user awareness of what the autonomy job entails to better inform the user if the autonomous operation should be commenced. The combination is further obvious since Ogura at least suggests that pathing information would be displayed along with the map data, despite the path not being shown in the drawings along with the map on the display map in work state display area 113c shown in various figures, i.e., 4 and corresponding descriptions; ¶ 53 the set travel route R is set previously in the field H as shown in Fig. 3).

In addition, Ogura in view of GoI and further in view of Foster fail to explicitly disclose the autonomy approval function providing a display indicative of at least one system check performed by the autonomous work machine.

Ikenori, from the same field of endeavor, discloses a mobile terminal with a display (3, Fig. 1-2) that displays an autonomy approval function that provides a display indicative of at

least one system check performed by the autonomous work machine (FIG. 15-22 and corresponding description, i.e., FIG. 21, “check complete”, FIG. 22 check completed . . . start autonomous drive 32d)

(¶ 148 As illustrated in FIG. 15, a home screen 32A of the liquid crystal monitor 32 displays a mode selection button (mode selector) 32a that enables selection of the autonomous drive mode and the like. When the autonomous drive mode is selected through operation of the mode selection button 32a, the autonomous drive control unit 22F performs a condition establishment determination process of determining whether or not all conditions for the transition of the drive mode of the tractor 1 from the manual drive mode to the autonomous drive mode are established. If all of the conditions are established, the autonomous drive control unit 22F causes the drive mode of the tractor 1 to transition from the manual drive mode to the autonomous drive mode, and notifies the mobile communication terminal 3 about the transition of the drive mode of the tractor 1 to the autonomous drive mode so as to enable the start of autonomous drive by operation of the mobile communication terminal 3.;

¶ 152 The execution of an initial check is premised on the establishment of all of first to ninth conditions described below (conditions for starting the operation check process);

¶ 191-192 The safety brake function unit 22Fa keeps the display of the completion of the initial check on the safety brake check screen 32C for a predetermined time in the initial check successful state. When the predetermined time elapses, the safety brake function unit 22Fa causes a transition to an initial check completed state (operation confirmed state) (“Check_OK” in FIG. 24), causes the display screen of the liquid crystal monitor 32 to transition to the autonomous drive start screen 32D illustrated in FIG. 22, and permits the transition of the drive mode from the manual drive mode to the autonomous drive mode. When various setting operations necessary for the autonomous drive of the tractor 1, such as the setting of the engine speed for autonomous drive through an operation of the accelerator bar and the setting of the vehicle speed for autonomous drive through an operation of the speed change lever, are

performed after the safety brake function unit 22Fa has caused a transition to the initial check completed state, all conditions for the transition of the drive mode to the autonomous drive mode are established. In a state in which all conditions are established, if an autonomous drive start button 32d displayed on the autonomous drive start screen 32D illustrated in FIG. 22 is operated, the autonomous drive control unit 22F causes the drive mode of the tractor 1 to transition from the manual drive mode to the autonomous drive mode, notifies the mobile communication terminal 3 about the transition of the drive mode of the tractor 1 to the autonomous drive mode, and enables the autonomous drive to be started by an operation of the mobile communication terminal 3.)

Accordingly, it would have been obvious to one of ordinary skill in the art at the time of effective filing date for the autonomy approval function in Ogura in view of Gol and further in view of Foster to provide a display indicative of at least one system check performed by the autonomous work machine, as disclosed by Ikenori, in order to improve safety by preventing automatic driving of the work vehicle while an error continues inside the vehicle (i.e., Ikenori, ¶ 28) and to prevent a decrease in work efficiency (Ikenori ¶202).

In addition, Ogura in view of Gol and further in view of Foster and further in view of Ikenori disclose a depiction of the autonomous work machine on the display

(Ogura: ¶¶ 84-85 “display device 113 . . . work state display area 113C . . . travel route R, an actual position, worked position and the like can be displayed”; 113C depicting icons of an autonomous work machine actual position, i.e., in FIG. 4-7, 9, etc.; 90 the remote control area 113A and the work state display area 113C may be displayed so that the autonomously moving work vehicle 1 can be operated while the work position is checked.”; Gol: 2, FIG. 6 and corresponding description; 12, FIG. 3-5 and corresponding description; Foster: 68-72, FIG. 3 and corresponding description) but fail to explicitly disclose displaying a depiction of the autonomous work machine with a visual indication that the autonomous work machine is initiating motion.

Fuji, from the same field of endeavor, discloses a display that provides a visually indicated alert that an autonomous vehicle will begin a particular motion.

(FIG. 5-7 and corresponding description; FIG. 8, "countdown"; FIG. 12 countdown; ¶¶ 63, 64, 66-67 countdown display part 51; 73-81; 84-87; 98-101; 106-116, etc., i.e., element 51 in FIG. 5-7 is a countdown that is a visual indication the autonomous work machine is initiating an autonomous motion. Element 251 in FIG. 14-16 is also a visual indication the autonomous work machine is initiating an autonomous motion; arrow 355 shown in FIG. 18 with display 314a/314b is a display of a depiction of the autonomous vehicle indicating when the AV will be initiating a motion as indicated by the position of the start of arrow 355 and as also indicated by countdown display 351. Arrow 355 is also shown in FIG. 8 (D) with vehicle A and its corresponding representation indicated by the arrow emerging from vehicle A (¶¶ 112-116))

Accordingly, it would have been obvious to one of ordinary skill in the art at the time of effective filing date to implement countdown timing display of Fuji into the system of Ogura in view of Gol and further in view of Foster and further in view of Ikenori in order to improve operator awareness and notice of impending autonomous movement, reduce uneasiness of the operator, to improve safety of autonomous operations and reduce operator uncertainty of impending autonomous actions (Fuji, 7, 11, 94-100; 110).

With respect to claim 2, Ogura in view of Gol and further in view of Foster and further in view of Ikenori and further in view of Fuji disclose the mobile device is a smartphone (Gol, ¶ 37 "interface on a remote device . . . smart phone . . . tablet")

With respect to claim 3, Ogura in view of Gol and further in view of Foster and further in view of Ikenori and further in view of Fuji disclose the mobile device is a tablet computer (Ogura, ¶ 31 remote control apparatus 112; 51 remote control apparatus 112 . . . can be taken out from the auxiliary moving work vehicle 100 and operated while being carried . . . tablet type

personal computer; 52 remote control apparatus 112 and the autonomously moving work vehicle 1 can be communicated with each other . . . display device 113 which is a touch panel-type operation screen which can be operated by touching the screen is provided; GoI, ¶ 37 "interface on a remote device . . . smart phone . . . tablet").

With respect to claim 10, Ogura in view of GoI and further in view of Foster and further in view of Ikenori and further in view of Fuji disclose the display provides a countdown timer indicating time until the autonomous work machine will begin motion.

(Fuji, FIG. 5-7 and corresponding description; FIG. 8, "countdown"; FIG. 12 countdown; ¶¶ 63, 64, 66-67 countdown display part 51; 73-81; 84-87; 98-101; 106-116, etc.)

With respect to claim 13, Ogura in view of GoI and further in view of Foster and further in view of Ikenori and further in view of Fuji disclose the autonomy management application provides an edit user interface element that is configured to receive user input and wherein the processor is configured to selectively allow the user to edit the autonomy job based on user input to the edit user interface element.

(Ogura, ¶ 57 change autonomous speed, change autonomous engine rotation speed; setting button; 58; 59; 72; 84).

With respect to claim 14, Ogura in view of GoI and further in view of Foster and further in view of Ikenori and further in view of Fuji disclose the begin mission user interface element requires the user to slide the user interface element.

(Ogura, ¶ 66 by sliding the automatic travel start button . . . shifted to automatic travel mode).

Claim 4 is rejected under 35 U.S.C. 103 as being unpatentable over Ogura in view of Gol and further in view of Foster and further in view of Ikenori and further in view of Fuji and further in view of US 20140028440 to Takeuchi et al. (Takeuchi)

With respect to claim 4, Ogura in view of Gol and further in view of Foster and further in view of Ikenori and further in view of Fuji disclose user proximity within the pre-determined distance is detected using EM radiation

(Gol, ¶ 39 citing various EM range detection sensors that ¶ 40 may be used to detect user proximity, further described in ¶ 42)

However, Ogura in view of Gol and further in view of Foster and further in view of Ikenori and further in view of Fuji fail to disclose the proximity detection is performed by comparing GPS coordinates of the autonomous work machine with GPS coordinates of the mobile device.

Takeuchi, from the same field of endeavor, discloses proximity detection of a user within a threshold distance can be carried out using EM radiation with either a proximity sensor or Bluetooth, as disclosed by Ogura, or additionally GPS can be used (¶ 20, claim 1, 104, FIG. 1, proximity between vehicle 100 and mobile device 110 and corresponding description; FIG. 2, step 202-204).

Accordingly, given the limited number of options for detecting proximity of a mobile device and a vehicle, it would have been an obvious design choice to select determining proximity by comparing GPS coordinates of a vehicle and GPS coordinates of the mobile device, as taught by Takeuchi. In addition, the modification provides the benefit of avoiding line of sight interference that would occur in some proximity detection means such as radar or infrared.

Claims 6-9 are rejected under 35 U.S.C. 103 as being unpatentable over Ogura in view of Gol and further in view of Foster and further in view of Ikenori and further in view of Fuji and further in view of US 20170079195 to Yokoyama (Yoko)

With respect to claim 6, Ogura in view of Gol and further in view of Foster and further in view of Ikenori and further in view of Fuji fail to explicitly disclose the display provides an indication of a perception system check.

Yoko, from the same field of endeavor, discloses displaying an indication of a perception system check (i.e., FIG. 5 S4, sensors have abnormality? S5 display; ¶ 71 In step S4, the sensors are checked (S4). It is checked whether in each of the sensors, a predetermined output voltage is outputted (an abnormal voltage is not outputted), a short circuit or disconnection is caused or not, and so on. If there is any abnormality, a sensor name and an abnormality content are displayed on the display 113 (S5), and the processing shifts to step S6. As the sensors are cited, for example, the steering sensor 20, the vehicle speed sensor 27, the level sensor 29, the gyro sensor 31, the azimuth sensor 32, an obstacle sensor 41, the camera 42, a position sensor that detects a speed change position of a main speed change lever, a speed change position of a sub-speed change lever, and a speed change position of a PTO speed change lever, a position sensor that detects a raising/lowering height of the rotary tilling device 24, a voltage sensing sensor of the battery and the like. As for a rotation sensor, a water temperature sensor, a hydraulic sensor and the like provided in the engine 3, if there is any abnormality, the abnormality is sensed and displayed as an abnormality of the engine through an engine controller 60. If there is no abnormality in the sensors, the processing also shifts to step S6.).

Accordingly, it would have been obvious to one of ordinary skill in the art at the time of effective filing date to implement the perception system check of Yoko into the system of Ogura in view of Gol and further in view of Foster and further in view of Ikenori and further in view of Fuji disclose in order to prevent accidents (Yoko, ¶ 18).

With respect to claim 7, Ogura in view of Gol and further in view of Foster and further in view of Ikenori and further in view of Fuji fail to disclose the display provides an indication of a positioning system check.

Yoko, from the same field of endeavor, discloses displaying an indication of a perception system check (i.e., FIG. 5 S8, satellite positioning has abnormality? S9 display; ¶¶ 73-76)

Accordingly, it would have been obvious to one of ordinary skill in the art at the time of effective filing date to implement the position system check of Yoko into the system of Ogura in view of Gol and further in view of Foster and further in view of Ikenori and further in view of Fuji in order to prevent accidents (Yoko, ¶ 18).

With respect to claim 8, Ogura in view of Gol and further in view of Foster and further in view of Ikenori and further in view of Fuji fail to explicitly disclose the display provides an indication of a tractor subsystem check.

Yoko, from the same field of endeavor, discloses displaying an indication of a tractor subsystem check (i.e., FIG. 5 S6, actuators have an abnormality? S7 display; ¶¶ 72-76; 25 swash plate; 27 steering cylinder; 31 tilling device; 43 raising/ lowering actuator; 50 axle actuator; 57 braking actuator; 69; claim 5)

Accordingly, it would have been obvious to one of ordinary skill in the art at the time of effective filing date to implement the tractor subsystem check of Yoko into the system of Ogura in view of Gol and further in view of Foster and further in view of Ikenori and further in view of Fuji in order to prevent accidents (Yoko, ¶ 18).

With respect to claim 9, Ogura in view of Gol and further in view of Foster and further in view of Ikenori and further in view of Fuji fail to explicitly disclose the display provides an indication of an implement subsystem check.

Yoko, from the same field of endeavor, discloses displaying an indication of an implement subsystem check (i.e., FIG. 5 S6, actuators have an abnormality? S7 display; ¶¶ 72-76; 25 swash plate; 27 steering cylinder; 31 tilling device; 43 raising/ lowering actuator; 50 axle actuator; 57 braking actuator; 69; claim 5)

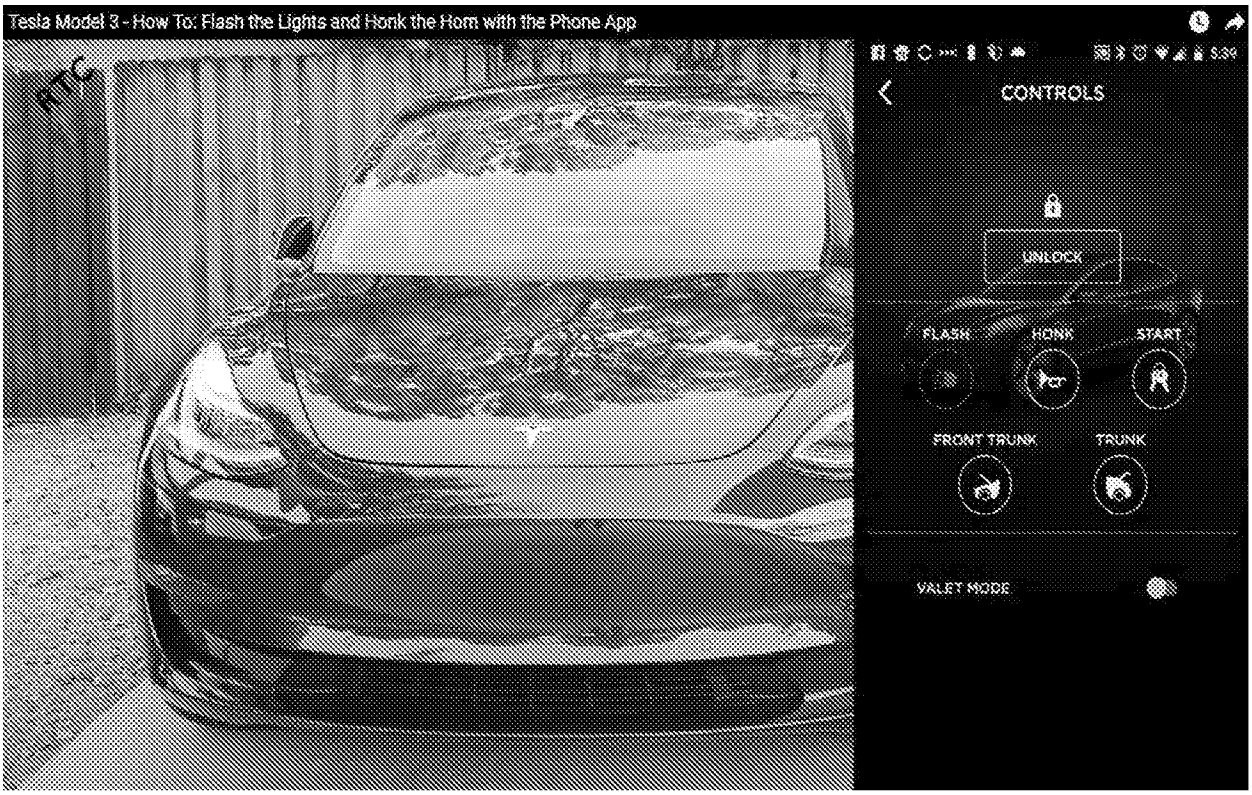
Accordingly, it would have been obvious to one of ordinary skill in the art at the time of effective filing date to implement the subsystem check of Yoko into the system of Ogura in view of Gol and further in view of Foster and further in view of Ikenori and further in view of Fuji in order to prevent accidents (Yoko, ¶ 18).

Claims 11-12 are rejected under 35 U.S.C. 103 as being unpatentable over Ogura in view of Gol and further in view of Foster and further in view of Ikenori and further in view of Fuji and further in view of Youtube publication "Tesla Model 3 – How to: Flash the lights and Honk the Horn with the App" to Cragun, published March 9, 2018, time mark 0:00 – 0:32 available at https://www.youtube.com/watch?v=JUb3kli3e0E (Cragun)

With respect to claim 11, Ogura in view of Gol and further in view of Foster and further in view of Ikenori and further in view of Fuji fail to explicitly disclose the display provides a flashing indication relative to lights of the autonomous work machine and also causes corresponding lights on the autonomous work machine to flash.

Cragun, from the same field of endeavor discloses a display that provides a flashing indication relative to lights of the autonomous work machine and also causes corresponding lights on the autonomous work machine to flash.

(time mark 0:03 – 0:30, see also screenshot below wherein the app display on a mobile phone flashes the "flash" icon while also causing lights on the autonomous vehicle to flash as shown in video)



Accordingly, it would have been obvious to one of ordinary skill in the art at the time of effective filing date to implement the teachings of Cragun into the system of Ogura in view of Gol and further in view of Foster and further in view of Ikenori and further in view of Fuji such that the display of Ogura in view of Gol and further in view of Foster and further in view of Ikenori and further in view of Fuji provides a flashing indication relative to lights of the autonomous work machine and also causes corresponding lights on the autonomous work machine to flash in order to provide increased user convenience so that the user can locate the autonomous work machine in order to find it and get within a predetermined distance to initiate autonomous activities.

With respect to claim 12, Ogura in view of Gol and further in view of Foster and further in view of Ikenori and further in view of Fuji and further in view of Cragun disclose the autonomy approval function provides a halt user interface element that is configured to receive user input

and wherein the processor is configured to selectively halt motion of the autonomous work machine based on user input to the halt user interface element.

(Ogura, ¶ 57, automatic travel stop button 204; 60 “the automatic travel and the work are stopped. By touching the sudden stop button 205, the engine is stopped so as to stop the travel and the work, thereby preventing dangerous state”).

Claims 15-17 and 20 are rejected under 35 U.S.C. 103 as being unpatentable over Ogura in view of Gol and further in view of Foster and further in view of Ikenori and further in view of Fuji and further in view of US 20170079195 to Yokoyama (Yoko)

With respect to claim 15, Ogura discloses a computer-implemented method of providing autonomy approval for an autonomous work machine, the method comprising:

displaying a selected field of an autonomy job for an autonomous work machine to a user;

receiving user approval of the autonomy job;

(¶ 31 remote control apparatus 112; 51 remote control apparatus 112 . . . can be taken out from the auxiliary moving work vehicle 100 and operated while being carried . . . tablet type personal computer; 52 remote control apparatus 112 and the autonomously moving work vehicle 1 can be communicated with each other . . . display device 113 which is a touch panel-type operation screen which can be operated by touching the screen is provided; 55 operator can operate the remote control apparatus 112 while supervising the autonomously moving work vehicle 1; 56-57 remote control area 113A is a screen in which operation members (buttons) for travel, steering and operation of the work machine of the autonomously moving work vehicle 1 are arranged . . . automatic travel start button 203, an automatic travel stop button 204, a sudden stop button 205; ¶¶ 64-67)

(64 automatic travel start button 203 . . . button for starting work while traveling automatically on the travel route R set previously . . . shifted to automatic travel mode . . . automatic travel control is performed by the control device 30)

(i.e., map in work state display area 113c shown in various figures, i.e., 4 and corresponding descriptions; ¶ 53 the set travel route R is set previously in the field H as shown in Fig. 3)

(mobile device 112, display 113 shown in FIG. 2, 4-12 and corresponding description; ¶¶ 57 -59 remote control area 113A, screen in which operation members (buttons) for travel, steering, an operation of the work machine of the autonomously moving work vehicle are arranged with various examples; 60-67; 71-74);

providing a begin mission user interface element configured to receive user input to selectively cause the autonomy approval function to initiate motion based on the user input to the begin mission user interface element; and

(i.e., ¶¶ 56-57 remote control area 113A is a screen in which operation members (buttons) for travel, steering and operation of the work machine of the autonomously moving work vehicle 1 are arranged . . . automatic travel start button 203, an automatic travel stop button 204, a sudden stop button 205; ¶¶ 60-67; ¶¶ 57 -59 remote control area 113A, screen in which operation members (buttons) for travel, steering, an operation of the work machine of the autonomously moving work vehicle are arranged with various examples)

However, Ogura fails to explicitly disclose the autonomy approval function being configured to instruct the user to move within a pre-determined distance from of an autonomous work machine that will execute the autonomy job, and provide a machine-verified indication of user proximity within the pre-determined distance such that the begin mission user interface is provided based on proximity of the user within the predetermined distance, i.e., an ability to select initiation of an autonomous motion can begin once the user is a predetermined distance from the work machine.

GoI is from the same field of endeavor, as GoI also discloses a remote device (i.e., 14, FIG. 1) wherein the remote device, like Ogura can be a tablet with a display (¶ 37 "interface on a remote device . . . smart phone . . . tablet") which allows a user to control motions of a vehicle in an unmanned manner (¶ 38 control the motive functions of the vehicle . . . without a driver occupying the vehicle) wherein GoI further discloses an autonomy approval function being configured to instruct the user to move within a pre-determined distance from of an autonomous work machine that will execute an autonomy job, and provide a machine-verified indication of user proximity within the pre-determined distance such that an automated job can only be performed after verification that a user has moved within the pre-determined distance of the autonomous work machine to initiate motion.

(¶¶ 37 system autonomously parks the vehicle 12 . . . user U may activate the remote parking system 10 using a button on a remote device 14 . . . smart phone . . . tablet, etc.; 41-42 remote device 14, the wireless communication circuit 24 facilitates communication between the user U and the system 10 (e.g., activating the remote parking system, etc.) . . . detect the location of the user U based on the range detection sensors . . . detect a location of the remote device 14 to position the user U . . . within a predetermined range of the vehicle 12 . . . as a condition to activating the remote parking routine; 48 require user is in viewing zone to initiate autonomous operation of the vehicle)

(¶ 37 interactive routine . . . instruct . . . a position of the user U within a viewing zone . . . guidance routine to position a user . . . in the viewing zone; 40 including references incorporated by reference; 49 user instructed to relocate within a predetermined distance, i.e., within viewing zone; 52-53, FIG. 4-6 including instructions on remote device; FIG. 8 142 "outputting instructions coaching the user to enter a viewing zone").

Accordingly, it would have been obvious to one of ordinary skill in the art at the time of effective filing date to implement the teachings of GoI into the system of Ogura such that a begin mission user interface element (i.e., Ogura start 203, FIG. 6, 203A, FIG. 9) is displayable after

verifying a user has moved withing a predetermined distance of an autonomous vehicle, as taught by GoI, since GoI discloses user proximity to the vehicle is a precondition to a user being able to approve initiation of motion in order to improve the ability of a user to monitor autonomy functions and in order to improve safety by ensuring the user does not interfere with the movement of the autonomous vehicle while also allowing the user to monitor operations of the AV, i.e., ¶ 1 "assisting a user-pedestrian monitoring the operation of a vehicle"; 37 the system still may benefit from pedestrian-user observation and monitoring . . . optimized viewing).

In addition, Ogura in view of GoI fail to explicitly disclose the display map includes displaying *pathing information* relative to the selected field, although Ogura at least suggests this feature (i.e., map in work state display area 113c shown in various figures, i.e., 4 and corresponding descriptions; ¶ 53 the set travel route R is set previously in the field H as shown in Fig. 3).

However, displaying pathing information along with map information was well known in the art at the time of effective filing. For example, Foster, from the same field of endeavor, discloses a user interface display (FIG. 3 and corresponding description) with a mapped grid of a field (right side, FIG. 3 including 74 (¶ 29 "agricultural field 74)) that includes pathing information (74, 76, FIG. 3 and corresponding description, ¶28 "controller may plan a path 76 for the agricultural vehicle . . . to traverse"). In addition, Foster discloses includes an autonomy approval function on the user interface to receive user acceptance of the job including a begin mission user interface element receiving user input to selectively cause initiation of motion (¶29 "controller may then receive a third input from the operator indicative of instructions to commence an autonomous agricultural operation on the agricultural field . . . touchscreen display 34 indicative of instructions to display a start button 78 while the virtual agricultural vehicle 70 and/or implement 72 and the virtual agricultural field 74 are selected. The controller may receive signal(s) from the display indicative of actuation of the start button 78 and may commence the autonomous agricultural operation accordingly (e.g., by sending signal(s) to the

steering control system and/or the speed control system indicative of instructions to direct the agricultural system along the path; 37; 39; claim 13).

Accordingly, it would have been obvious to one of ordinary skill in the art at the time of effective filing date to include pathing information in the user display as disclosed by Foster above, in the system of Ogura in view of Gol in order to improve user awareness of what the autonomy job entails to better inform the user if the autonomous operation should be commenced. The combination is further obvious since Ogura at least suggests that pathing information would be displayed along with the map data, despite the path not being shown in the drawings along with the map on the display map in work state display area 113c shown in various figures, i.e., 4 and corresponding descriptions; ¶ 53 the set travel route R is set previously in the field H as shown in Fig. 3).

In addition, Ogura in view of Gol and further in view of Foster fail to explicitly disclose the step of executing a system check on the autonomous work machine based on user approval of the autonomy job.

Ikenori, from the same field of endeavor, discloses a mobile terminal with a display (3, Fig. 1-2) that displays an autonomy approval function that facilitates executing a system check on the autonomous work machine based on user approval of the autonomy job

(FIG. 15-22 and corresponding description, i.e., FIG. 21, “check complete”, FIG. 22 check completed . . . start autonomous drive 32d)

(¶ 148 As illustrated in FIG. 15, a home screen 32A of the liquid crystal monitor 32 displays a mode selection button (mode selector) 32a that enables selection of the autonomous drive mode and the like. When the autonomous drive mode is selected through operation of the mode selection button 32a, the autonomous drive control unit 22F performs a condition establishment determination process of determining whether or not all conditions for the transition of the drive mode of the tractor 1 from the manual drive mode to the autonomous drive mode are established. If all of the conditions are established, the autonomous drive control unit

22F causes the drive mode of the tractor 1 to transition from the manual drive mode to the autonomous drive mode, and notifies the mobile communication terminal 3 about the transition of the drive mode of the tractor 1 to the autonomous drive mode so as to enable the start of autonomous drive by operation of the mobile communication terminal 3.;

¶ 152 The execution of an initial check is premised on the establishment of all of first to ninth conditions described below (conditions for starting the operation check process);

¶ 191-192 The safety brake function unit 22Fa keeps the display of the completion of the initial check on the safety brake check screen 32C for a predetermined time in the initial check successful state. When the predetermined time elapses, the safety brake function unit 22Fa causes a transition to an initial check completed state (operation confirmed state) ("Check_OK" in FIG. 24), causes the display screen of the liquid crystal monitor 32 to transition to the autonomous drive start screen 32D illustrated in FIG. 22, and permits the transition of the drive mode from the manual drive mode to the autonomous drive mode. When various setting operations necessary for the autonomous drive of the tractor 1, such as the setting of the engine speed for autonomous drive through an operation of the accelerator bar and the setting of the vehicle speed for autonomous drive through an operation of the speed change lever, are performed after the safety brake function unit 22Fa has caused a transition to the initial check completed state, all conditions for the transition of the drive mode to the autonomous drive mode are established. In a state in which all conditions are established, if an autonomous drive start button 32d displayed on the autonomous drive start screen 32D illustrated in FIG. 22 is operated, the autonomous drive control unit 22F causes the drive mode of the tractor 1 to transition from the manual drive mode to the autonomous drive mode, notifies the mobile communication terminal 3 about the transition of the drive mode of the tractor 1 to the autonomous drive mode, and enables the autonomous drive to be started by an operation of the mobile communication terminal 3.).

Accordingly, it would have been obvious to one of ordinary skill in the art at the time of effective filing date for the autonomy approval function in Ogura in view of Gol and further in view of Foster to execute a system check on the autonomous work machine based on user approval of the autonomy job, as disclosed by Ikenori, in order to improve safety by preventing automatic driving of the work vehicle while an error continues inside the vehicle (i.e., Ikenori, ¶ 28) and to prevent a decrease in work efficiency (Ikenori ¶202).

In addition, Ogura in view of Gol and further in view of Foster and further in view of Ikenori disclose a depiction of an autonomous work machine on the display (Ogura: ¶¶ 84-85 "display device 113 . . . work state display area 113C . . . travel route R, an actual position, worked position and the like can be displayed"; 113C depicting icons of an autonomous work machine actual position, i.e., in FIG. 4-7, 9, etc.; 90 the remote control area 113A and the work state display area 113C may be displayed so that the autonomously moving work vehicle 1 can be operated while the work position is checked."; Gol: 2, FIG. 6 and corresponding description; 12, FIG. 3-5 and corresponding description; Foster: 68-72, FIG. 3 and corresponding description) but fail to explicitly disclose generating an alert that motion startup is occurring wherein the alert includes a visual indication that the autonomous work machine is initiating motion.

Fuji, from the same field of endeavor, discloses generating an alert including a depiction with a visual indication that the autonomous work machine is initiating motion

(FIG. 5-7 and corresponding description; FIG. 8, "countdown"; FIG. 12 countdown; ¶¶ 63, 64, 66-67 countdown display part 51; 73-81; 84-87; 98-101; 106-116, etc., i.e., element 51 in FIG. 5-7 is a countdown that is a visual indication the autonomous work machine is initiating an autonomous motion. Element 251 in FIG. 14-16 is also a visual indication the autonomous work machine is initiating an autonomous motion; arrow 355 shown in FIG. 18 with display 314a/314b is a display of a depiction of the autonomous vehicle indicating when the AV will be initiating a motion as indicated by the position of the start of arrow 355 and as also indicated by

countdown display 351. Arrow 355 is also shown in FIG. 8 (D) with vehicle A and its corresponding representation indicated by the arrow emerging from vehicle A (¶¶ 112-116))

Accordingly, it would have been obvious to one of ordinary skill in the art at the time of effective filing date to implement the generated alert of Fuji into the system of Ogura in view of Gol and further in view of Foster and further in view of Ikenori in order to improve operator awareness and notice of impending autonomous movement, reduce uneasiness of the operator, to improve safety of autonomous operations and reduce operator uncertainty of impending autonomous actions (Fuji, 7, 11, 94-100; 110).

In addition, Ogura in view of Gol and further in view of Foster and further in view of Ikenori and further in view of Fuji fail to explicitly disclose the system check is a perception system check.

Yoko, from the same field of endeavor, discloses displaying an indication of a perception system check (i.e., FIG. 5 S4, sensors have abnormality? S5 display; ¶ 71 In step S4, the sensors are checked (S4). It is checked whether in each of the sensors, a predetermined output voltage is outputted (an abnormal voltage is not outputted), a short circuit or disconnection is caused or not, and so on. If there is any abnormality, a sensor name and an abnormality content are displayed on the display 113 (S5), and the processing shifts to step S6. As the sensors are cited, for example, the steering sensor 20, the vehicle speed sensor 27, the level sensor 29, the gyro sensor 31, the azimuth sensor 32, an obstacle sensor 41, the camera 42, a position sensor that detects a speed change position of a main speed change lever, a speed change position of a sub-speed change lever, and a speed change position of a PTO speed change lever, a position sensor that detects a raising/lowering height of the rotary tilling device 24, a voltage sensing sensor of the battery and the like. As for a rotation sensor, a water temperature sensor, a hydraulic sensor and the like provided in the engine 3, if there is any abnormality, the abnormality is sensed and displayed as an abnormality of the engine through an engine controller 60. If there is no abnormality in the sensors, the processing also shifts to step S6.).

Accordingly, it would have been obvious to one of ordinary skill in the art at the time of effective filing date to implement the perception system check of Yoko into the system of Ogura in view of Gol and further in view of Foster and further in view of Ikenori and further in view of Fuji disclose in order to prevent accidents (Yoko, ¶ 18).

With respect to claim 16, Ogura in view of Gol and further in view of Foster and further in view of Ikenori and further in view of Fuji and further in view of Yoko disclose user proximity within the pre-determined distance is detected using electromagnetic radiation.

(Gol, ¶ 39 citing various EM range detection sensors that ¶ 40 may be used to detect user proximity, further described in ¶ 42)

With respect to claim 17, Ogura in view of Gol and further in view of Foster and further in view of Ikenori and further in view of Fuji and further in view of Yoko disclose the begin mission user interface element is provided on a mobile device.

(Ogura, ¶ 31 remote control apparatus 112; 51 remote control apparatus 112 . . . can be taken out from the auxiliary moving work vehicle 100 and operated while being carried . . . tablet type personal computer; 52 remote control apparatus 112 and the autonomously moving work vehicle 1 can be communicated with each other . . . display device 113 which is a touch panel-type operation screen which can be operated by touching the screen is provided; Gol, ¶ 37 “interface on a remote device . . . smart phone . . . tablet”).

With respect to claim 20, Ogura in view of Gol and further in view of Foster and further in view of Ikenori and further in view of Fuji and further in view of Yoko disclose the autonomy approval function provides a halt user interface element that is configured to receive user input and wherein the processor is configured to selectively halt motion of the autonomous work machine based on user input to the halt user interface element.

(Ogura, ¶ 57, automatic travel stop button 204; 60 "the automatic travel and the work are stopped. By touching the sudden stop button 205, the engine is stopped so as to stop the travel and the work, thereby preventing dangerous state").

Conclusion

Applicant's amendment necessitated the new ground(s) of rejection presented in this Office action. Accordingly, **THIS ACTION IS MADE FINAL**. See MPEP § 706.07(a). Applicant is reminded of the extension of time policy as set forth in 37 CFR 1.136(a).

A shortened statutory period for reply to this final action is set to expire THREE MONTHS from the mailing date of this action. In the event a first reply is filed within TWO MONTHS of the mailing date of this final action and the advisory action is not mailed until after the end of the THREE-MONTH shortened statutory period, then the shortened statutory period will expire on the date the advisory action is mailed, and any nonprovisional extension fee (37 CFR 1.17(a)) pursuant to 37 CFR 1.136(a) will be calculated from the mailing date of the advisory action. In no event, however, will the statutory period for reply expire later than SIX MONTHS from the date of this final action.

Any inquiry concerning this communication or earlier communications from the examiner should be directed to KENNETH J MALKOWSKI whose telephone number is (313)446-4854. The examiner can normally be reached 8:00 AM - 5:00 PM.

Examiner interviews are available via telephone, in-person, and video conferencing using a USPTO supplied web-based collaboration tool. To schedule an interview, applicant is encouraged to use the USPTO Automated Interview Request (AIR) at http://www.uspto.gov/interviewpractice.

If attempts to reach the examiner by telephone are unsuccessful, the examiner's supervisor, Faris Almatrahi can be reached on 313-446-4821. The fax phone number for the organization where this application or proceeding is assigned is 571-273-8300.

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